

Right Whale Recognition

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I. BACKGROUND

The project is based on the Kaggle's \$10,000 competition in 2015 [1]. As there are less than 500 North Atlantic right whales in the world's oceans, it is essential for researchers to know their status in order to protect them from extinction. In addition, it is hard and inefficient for most researchers to identify single whale for biological samples, acoustic recordings, and necessary health assessments.

Right whales most distinguishing feature is the callosity pattern on top of a whale's head. Figure 6 shows the two types of callosity pattern: continuous and broken.

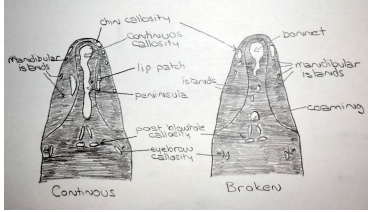


Fig. 1. Callosity pattern.

In the project, our challenge is to automate the right whale recognition process using a dataset of aerial photographs of individual whales. It would allow researchers to better focus on their conservation efforts. We will use MATLAB for this project and use some machine learning toolboxes.

II. DATA DESCRIPTION

There are 11458 images of North Atlantic right whales on Kaggle. These images are clean with only one whale in each photograph. But there are only 4544 images given with corresponding whale ids. Hence, the dataset of the project is the 4544 images. There are 447 unique whale ids.

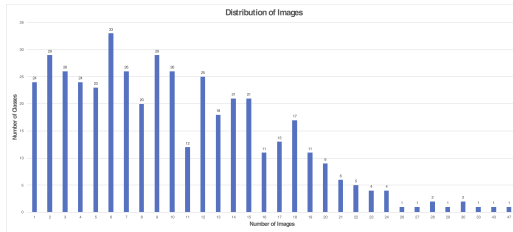


Fig. 2. Distribution of images

Figure 2 shows the distribution of the 4544 images. The x axis presents the number of images, and the y axis presents

the number of classes. The dataset is totally imbalanced. There are 447 classes, which means 447 unique whale ids in total. However, 24 classes have only one image and 111 classes with more than 15 images. The imbalance of the dataset and poor number of images in many classes are two big difficulties for recognition.

Another challenge is the qualities of images. As shown in Figure 2, blurred images, low light images, wave effected images make it hard to recognize. In addition, the photography distances and whale's activities make a big difference of images of a same whale. And for detection part, all images are manually rotated to east or north-east direction.

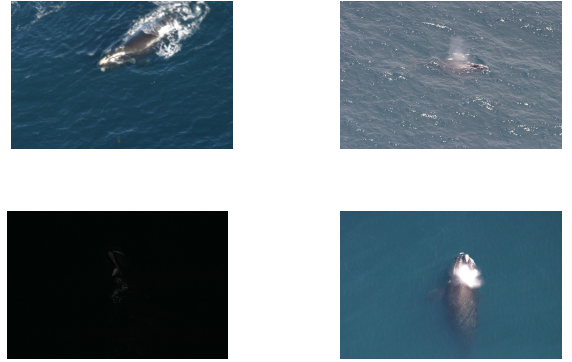


Fig. 3. Samples

In our project, the 4544 images are divided into subfolders according to whale ids based on the given csv file.

III. DETECTION

In matlab, there is a function called CascadeObjectDetector provided by Computer Vision System Toolbox. It uses the Viola-Jones algorithm to detect people's faces, noses, eyes, mouth or upper body. It can be easily implemented using 'Image Labeler' application which helps to set the ROI(Region of Interest) in each image. To use the CascadeObjectDetector, we need a few inputs.

A. Images to train

- Among the 4544 images, 1037 images are randomly selected to train and labeled noting the bounding box around the whale head to use as positive examples. For negative examples, by setting a fixed part of image and then cropping and storing, the water-only part of the image can be got.



Fig. 4. ImageLabeler application

B. Number of stages

- Looking at the structure of CascadeObjectClassifier, it consists of stages where stage is an ensemble of weak learners. Each stage of the classifier labels the region defined by the current location of the sliding window as either positive or negative. Positive indicates that an object was found and negative indicates no objects were found. It's not always the best idea to increase the number of stages. This increase results in a greater chance of rejecting a positive sample by mistake. So, we need to set the number of stages considerably considering the amount of training samples.

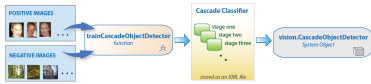


Fig. 5. Process of CascadeObjectDetector

C. Feature Type

- There are three kinds of features types that we can choose. Haar-like, Local Binary Patterns(LBP), Histogram of oriented gradients(HOG). Since HOG features are often used to detect the overall shape of an object, it wasn't appropriate. So I tried with two feature type; Haar and LBP. LBP feature type was much better as theoretically it's tolerant to the light compare to Haar-like.

Actually, the detector trained using CascadeObjectDetector is very sensitive to out-of-plane rotation. But whales can float on the ocean in any direction. So basically, training detector for every orientation(north, south, west, east) is necessary. However, time wasn't that enough and to make it simple, we manually changed all the orientation of the whale to make them see the same direction. This approach gave us comparatively good results for most of the images.

D. Control the number of bounding boxes

- Actually, the detection result is not only affected by some parameters like feature type, false positive rate, number of stages etc, but also the size of image. When the same image is tested for whale head detection, the result as image size varies. So, the procedure to find the most appropriate size was necessarily needed. The truth

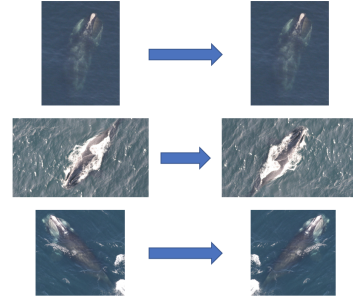


Fig. 6. Change the whale head direction

through many trials if the image is horizontal, it shows the best result when its size is 0.3 times of the original image size. In case of vertical image, best result is shown at the image size of 0.3 times of the original image size. That's the first procedure to find the optimal image size. Next step is also about the image resizing considering the number of bounding boxes(bbox). Making a bbox_history array and store the number of bbox in that array everytime the while loop iterates. Through this whole procedure, we can find most appropriate bounding box for the image.

It showed an accuracy of about 80% or more in the set of images with aligned head.

IV. RECOGNITION

Traditional machine learning techniques and deep learning techniques are applied in this section.

A. Data preprocessing

- In order to do recognition focus on whale heads, whale heads in the 4544 images are manually labeled with 'Image Labeler' and cropped.
- The dataset is randomly divided into train set (80%) and test set (20%).
- The train set is divided into two datasets: the new train set and validation set. The two new datasets both include all classes with one images. For classes with two to five images, only one image is taken out for validation set, the rest images are categorized into new train set. For the rest of the classes, images are divided randomly with 90% for new train set and 10% for validation set.
- As shown in Figure 2, 234 classes with less than 10 images. In order to increase the dataset, the following methods are used to oversampling aggressively. First is two contrast enhancement methods. As shown in Figure 7, unsharp filter can make the slightly blurred image more clear and in Figure 8, imadjust filter can make the highly blurred image more clear. Then, images are rotated, sheared and combined the two. Figure 9 shows all the nine oversampled images for a image. The default

oversampling has been done is to make nine copies of all images in the new train set.

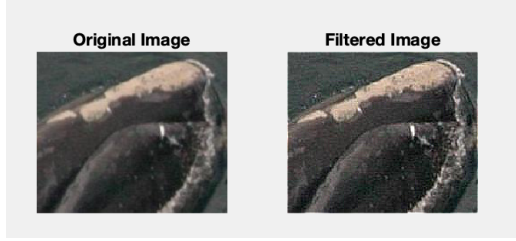


Fig. 7. Using unsharp contrast enhancement filter.



Fig. 8. Using imadjust contrast enhancement filter.

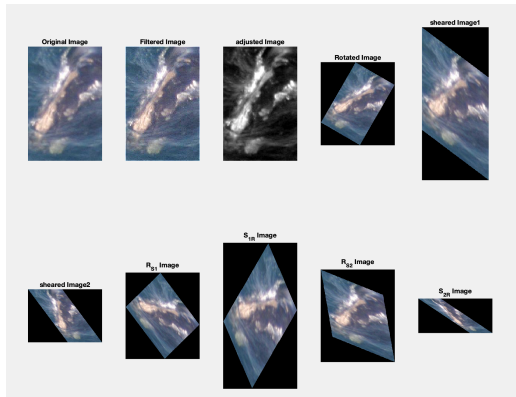


Fig. 9. All oversampled images.

B. Traditional machine learning

Traditional machine learning classification techniques for images contains two steps: feature extraction and classifier training. In this section, all images are resized to be 128×128 and turned to be gray scale.

- The first tried feature extraction method is hog features. As the key to distinguish whale is the shape of callosity pattern, Figure 10 indicates that cell size 8×8 is better than others to represent the shape. Thus, it is chosen and extracted 8100 features for a image. Then PCA is applied. Hog features are represented as principle components scores. In addition, only principle components scores which explained 95% variations are selected for classification. The feature size decreased from 8100 to about 1300 according to different training

set.

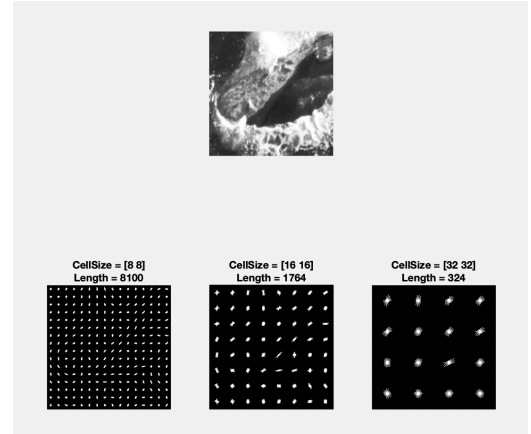


Fig. 10. using different cell size.

- The second tried feature extraction method is surf. Unlike hog feature extraction, images with same size can have different number of surf descriptors. Bag of features which also is referred to bag of words in MATLAB is convenient to use to capture strongest surf features for each class. Firstly, it extracts surf features for all images. Then 80% strongest features for each class are stored. Lastly, 500 words or features are generated as bag for classification using k-means clustering.
- With extracted features, linear svm is applied for classification. As the original dataset is highly imbalanced. The penalty factor 'C' which refers to 'BoxConstraint' in MATLAB is the hyperparameter used to tune the classifier. The range of 'C' is set as $[1e-3, 1e3]$. Then Bayesian optimization method is used to tune 'C', which is shown to be faster than grid search [2].
- For both svm with hog features and svm with bag of features, two kind of train set are used in the experiments. One is the original new train set. Another one is combination of the new train set and the oversampled images for classes with less than 10 images.
- The table below shows the accuracy for the two train set and the two kinds of feature extractions. The best accuracy is only 11.21%. Thus we did deep learning which will disussed int the next section.

TABLE I
RESULTS

Dataset	Feature Extraction type	Accuracy for test set
Original	hog + linear svm	2.87%
Oversampled	hog + linear svm	2.21%
Original	BoF + linear svm	6.19%
Oversampled	BoF + linear svm	11.21%

C. Deep learning

The most important challenge of this project is the data. We checked one of our model and measure the performance in different stage of classes (1.classes with more than 15 images in each class; 2. all classes) At last the performance in stage 1 is around 80%(depends on different network), while the accuracy of all classes are quite low.

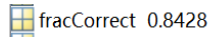


Fig. 11. The accuracy of stage 1

- Denoising. The model we mentioned before is trained for the whole dataset. Actually we can regard classes with only 1 image as noises to get higher accuracy. If we analyse the dataset, there are 24 classes with only 1 image. We can regard them as noise and retrain the network.
- Augmentation. The most suitable range of rotation in our case have been tried is [-10 10], and rescaling with the range from 0.9 to 1.1. In addition, XY-Reflection is also helpful. We also did some rotation and reflection manually to keep all the whale head points to the same direction, by doing that, the performance can be increased be around 5%.
- Comparison of different deep learning network This table shows a general difference between these 3 deep learning networks using in this project.

Model	AlexNet	VGG16	GoogLeNet
#Layers	8	16	22
Inception	-	-	+
#Convolution Layers	5	13	21
#Fullyconnected Layers	3	3	1
Dropout	+	+	+
Batch Normalization	-	-	-
Results	20-25%	25-40%	20%-80%

- Discussion. The results in the table above shows the accuracy range with different learning rates.

AlexNet is the most common Network, the advantage of AlexNet is we have Dropout Layer and using ReLU to make it runs quicker, so I can train this network with computer having no GPU. The result of it is not good it is around 20-25% accuracy.

VGG16 is Inheritance of AlexNet but much more deeper than AlexNet, structure of it is really better. The inputsize of each layer is from 224 to 7 (Each time multiply with 2) and the filter is from 64 to 512. And there are 5 Maxpooling Layers so it simplify the calculation, to help it quicker to get results. The result of VGG 16 is little better around 25-40% Sometimes if we don't give a suitable hyper parameter or did terrible

in data Augmentation the result can be even worse than AlexNet. That means only make our network deeper is not enough for our project. A deeper and bigger Network is hard to apply. And there are too many parameter in Network so it leads to Overfitting. Inception is a method to solve these issues.

GoogLeNet has the advantage. Even GoogLeNet is deeper than others it also shows better performance. Actually in real cases the performance floats form 20% to 80%.

Inception V3 which has the advantage: Factorization, it makes $7*7$ divide by 2 1 dimensional convolution ($1*7, 7*1$), and also the same thing to $3*3$, so it makes the network deeper and make the calculation quicker, and the input size is $299*299*3$. What's more, it has Batch Normalization, so we can have a higher dropout layer, use bigger learning rate. We have less Epoch in GoogLeNet and inception V3 than AlexNet and VGG16, so we can show what the image more exactly than AlexNet and VGG16.

Because our google cloud server shut downm, we lost some trained models, you can download one of our models with this link: <https://drive.google.com/drive/folders/1C-rvxVSaWiDmUvqqC16sOAEctvfkby?usp=sharing>

D. Conclusion

In conclusion, this project is quite challenge. More precise preprocessing methods needed for the recognition to focus on the callosity pattern of whale.

Traditional machine learning methods did not work well in this project. Since we do the deep learning using wholly manually labeled whale head (which ignored the results of the detection), so the real performance can be worse.

There are also differences between neural networks. Not the deeper the better. If we have bad input parameter, even we have a suitable network it won't give a good performance.

REFERENCES

- [1] NOAA FISHERIES (2015). Right Whale Recognition. Retrieved from <https://www.kaggle.com/c/noaa-right-whale-recognition>.
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